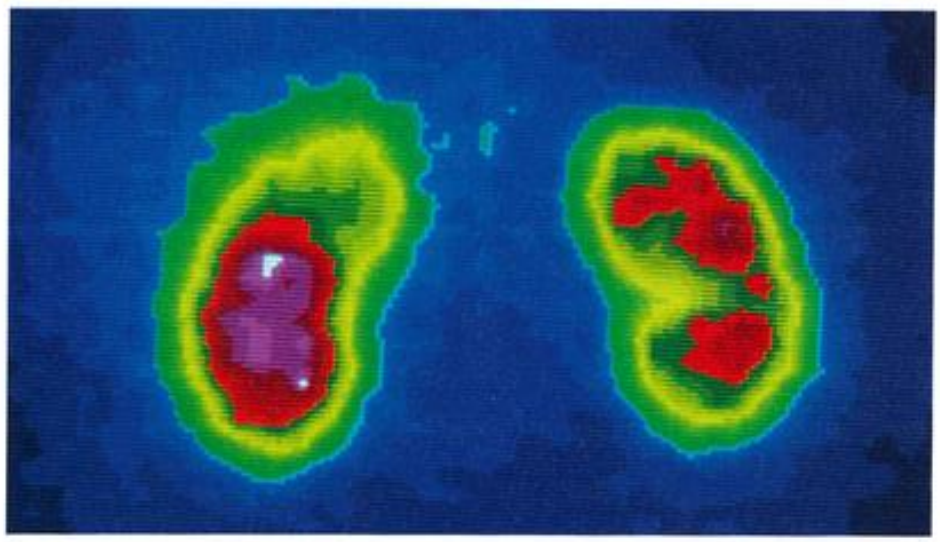
24 APPLICATIONS OF RADIOACTIVITY

Radioactivity has a wide variety of uses, including medicine, industry and power generation. In this chapter, you will read about these uses and also learn about the dangers associated with the use of radioactivity.

LEARNING OBJECTIVES

- Describe uses of radioactivity in industry and medicine
- Describe the difference between contamination and irradiation
- Be able to describe the dangers of ionising radiations including:
  - that radiation can cause mutations in living organisms
  - that radiation can damage cells and tissue
  - the problems arising from the disposal of radioactive waste and how the associated risks can be reduced

THE USE OF RADIOACTIVITY IN MEDICINE



▲ Figure 24.1 This scan shows the kidneys in a patient's body.

USING TRACERS IN DIAGNOSIS

Radioactive isotopes are used as tracers to help doctors identify diseased organs (like the kidneys or the liver). A radioactive tracer is a chemical compound that emits gamma radiation. The tracer is taken orally (swallowed) by the patient or injected. Its journey around the body can then be traced (followed) using a gamma ray camera.

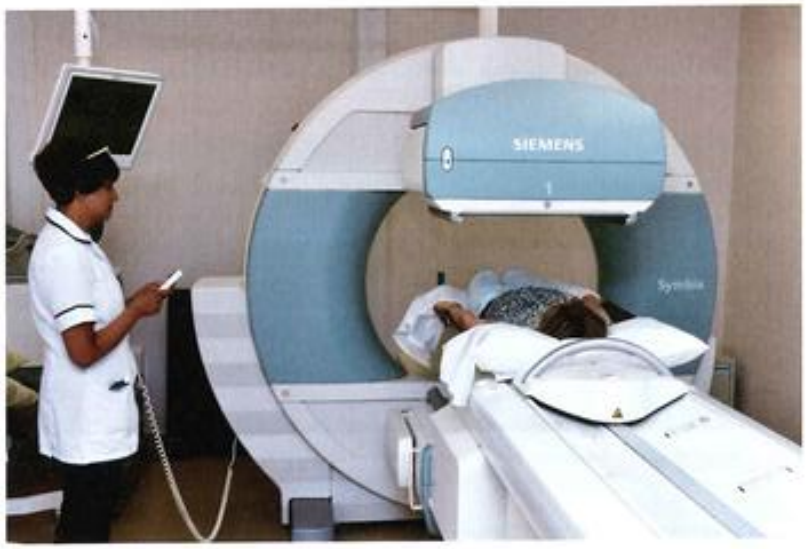
Different compounds are chosen for different diagnostic tasks. For example, the isotope iodine-123 is absorbed by the thyroid gland (a part of the body found in the neck that controls growth) in the same way as the stable form of iodine. The isotope decays and emits gamma radiation. A gamma ray camera can then be used to form a clear image of the thyroid gland.

The half-life of iodine-123 is about 13 hours. A short half-life is important as this means that the activity of the tracer decreases to a very low level in a few days.

EXTENSION WORK

The isotope used in medicine is what is called a metastable form of technetium-99 called technetium-99m.

Other isotopes are used to image specific parts of the body. Technetium-99 is the most widely used isotope in medical imaging. It is used to help identify medical problems that affect many parts of the body. Figure 24.1 shows a scan of a patient's kidneys. It shows clearly that one of the kidneys is not working properly.



▲ Figure 24.2 Scanner used to provide 3D images of a patient's body

Imaging techniques enable doctors to produce three-dimensional (3D) computer images of parts of a patient's body. These are of great value in diagnosis. Figure 24.2 shows the kind of equipment used for three-dimensional imaging.

TREATMENT

Radiation from isotopes can have various effects on the cells that make up our bodies. Low doses of radiation may have no lasting effect. Higher doses may cause the cells to stop working properly as the radiation damages the DNA in the cells. This can lead to abnormal growth and cancer. Very high doses will kill living cells.

Cancer can be treated by surgery that involves cutting out cancerous cells. Another way of treating cancer is to kill the cancer cells inside the body. This can be done with chemicals containing radioactive isotopes. Unfortunately, the radiation kills healthy cells as well as diseased ones. To reduce the damage to healthy tissue, chemicals are used to target the location of the cancer in the body. They may emit either alpha or beta radiation. Both these types of radiation have a short range in the body, so they will affect only a small volume of tissue close to the target.

The radioisotope iodine-131 is used in the treatment of various diseases of the thyroid gland. It has a half-life of about eight days and decays by beta particle emission.

STERILISATION USING RADIATION

Gamma radiation can kill **bacteria** and viruses. It is therefore used to kill these microorganisms on surgical instruments and other medical equipment. The technique is called irradiation. The items to be **sterilised** are placed in secure bags to ensure that they cannot be re-contaminated before use. The gamma radiation will pass through the packaging and destroy bacteria without damaging the item.

KEY POINT

Irradiation will not destroy any poisons that bacteria may have already produced in the food before it is treated.
Irradiation does not destroy vitamins in the food like other means of killing bacteria, such as high-temperature treatment.

Some food products are treated in a similar way to make sure that they are free from any bacteria that will cause the food to rot or will cause food poisoning. The irradiation of food is an issue that causes concern amongst the public and is not a widely used procedure at the present time.

Irradiation such as the deliberate exposure of food products and surgical instruments to controlled amounts of radiation should not be confused with radioactive contamination. If radioactive waste is accidentally released either into the air or the sea it could result in fish, animals or agricultural crops being contaminated with radioactive material.



▲ Figure 24.3 Gamma radiation is used to sterilise medical equipment.

THE USE OF RADIOACTIVITY IN INDUSTRY

GAMMA RADIOGRAPHY



▲ Figure 24.4 A gamma camera image used to view inside a valve

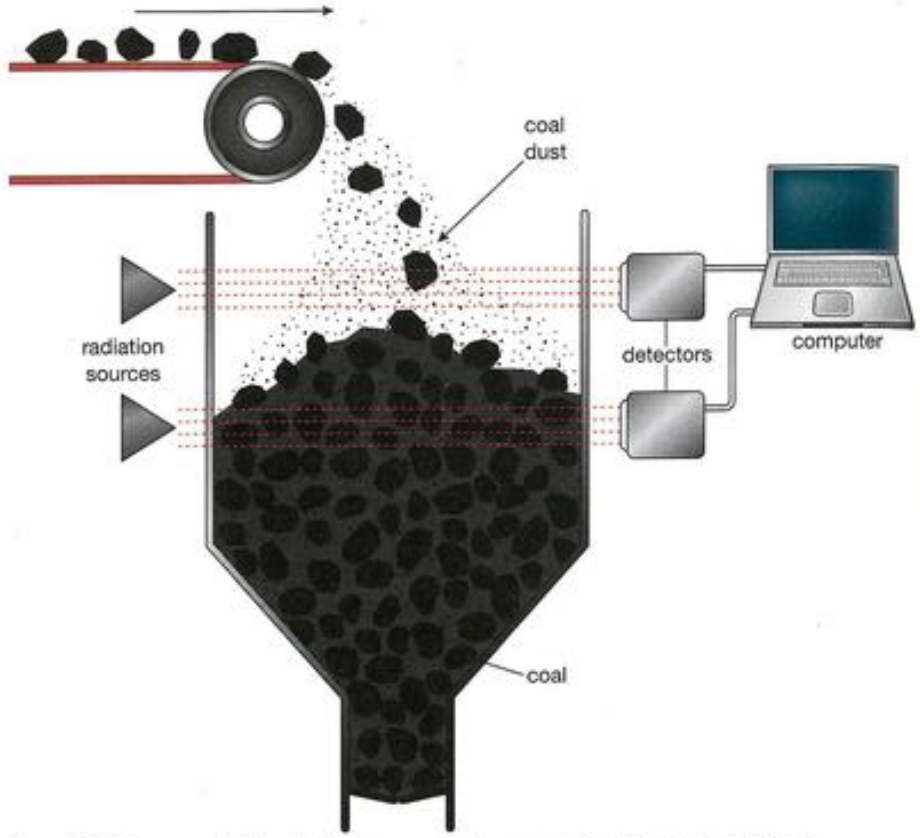
A gamma ray camera is like the x-ray cameras used to examine the contents of your luggage at airports. A source of gamma radiation is placed on one side of the object to be **scanned** and a gamma camera is placed on the other. Gamma rays pass through more objects than x-rays. They can be used to check for faults in casting (making things out of metal) or welding (joining metal objects together). Without the technique of gamma radiography, neither problem could be detected unless the welding or casting were cut through. An additional advantage of gamma radiography over the use of x-rays for this purpose is that gamma sources can be small and do not require a power source or large equipment.

GAUGING

In industrial processes, raw materials and fuel are stored in large tanks called hoppers. Figure 24.5 shows how radioactive isotopes are used to gauge, or measure, how much material there is in a storage container.

The coal absorbs a large amount of the radiation so the reading on the lower detector will be small. As the upper part of the hopper is empty the upper detector will have a high reading.

This method of gauging has several advantages over other methods. There is no contact with the material being gauged. Also, coal dust might cause false readings with an optical gauging system (one using light beams). Coal dust is much less dense than coal so the gamma ray system still works properly.

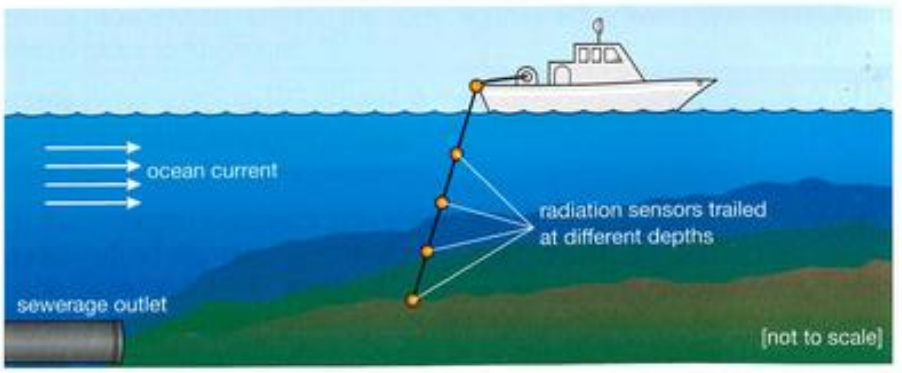


▲ Figure 24.5 The amount of coal in the hopper can be measured using gamma radiation.

Another example of gauging uses a similar process to monitor the thickness of plastic sheeting and film. The thicker the sheet, the greater the amount of radiation it absorbs and the amount passing through gets smaller. By measuring the amount of radiation that passes through the sheeting, its thickness can be closely controlled during manufacture.

TRACING AND MEASURING THE FLOW OF LIQUIDS AND GASES

Radioisotopes are used to check the flow of liquids in industrial processes. Very tiny amounts of radiation can easily be detected. Complex piping systems, like those used in in power stations, can be monitored for leaks. Radioactive tracers are even used to measure the rate of spread of sewage (human waste) (Figure 24.6)!



▲ Figure 24.6 Radioactive tracers released with the sewage allow its spread to be monitored to make sure the concentration does not reach harmful levels in any area.

RADIOACTIVE DATING

A variety of different methods involving radioisotopes are used to date minerals and organic matter. The most widely known method is radiocarbon dating. This is used to find the age of organic matter – for example, from trees and animals – that was once living. We shall also look at techniques that are used to find the age of inorganic material like rocks and minerals.

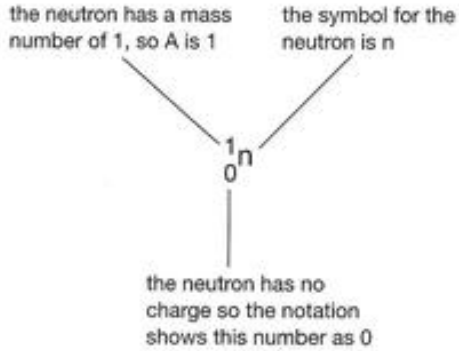
RADIOCARBON DATING

Radiocarbon dating measures the level of an isotope called carbon-14 (C-14). This is made in the atmosphere. Cosmic rays from space are continually raining down upon the Earth. These have a lot of energy. When they hit atoms of gas in the upper layers of the atmosphere, the nuclei of the atoms break apart. The parts fly off at high speed. If they hit other atoms they can cause nuclear transformations (changes) to take place. These transformations turn the elements in the air into different isotopes. One such collision involves a fast-moving neutron striking an atom of nitrogen. (Nitrogen forms nearly 80% of our atmosphere.) The nuclear equation for this process is:

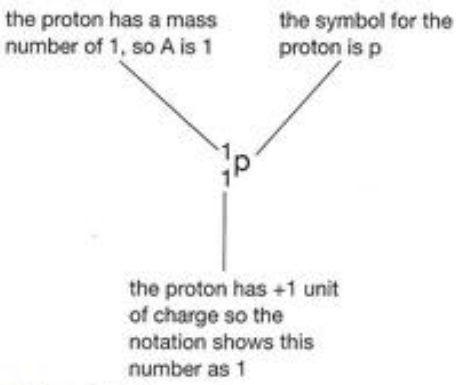
$$^{14}_{7}N + {}^{1}_{0}n \rightarrow {}^{14}_{6}C + {}^{1}_{1}p$$

Notice that, as in the other nuclear equations we have seen, the top numbers – which show the number of nucleons – add up to the same total on each side of the equation. This is because the mass number is conserved. The bottom numbers – which show the amount of charge on the particles – are also conserved.

You have seen the notation for neutrons and protons in Chapter 22. A reminder of neutron notation is shown in Figure 24.7, and of proton notation in Figure 24.8.



▲ Figure 24.7 The neutron



▲ Figure 24.8 The proton

The result of the collision of a neutron with a nitrogen atom is a nuclear transformation. The nitrogen atom is transformed into an atom of the radioactive isotope of carbon, carbon-14.

As we have already mentioned, isotopes of an element have the same chemical behaviour. This means that the carbon-14 atoms react with oxygen in our atmosphere to form carbon dioxide, just like the much more common and stable isotope, carbon-12. The carbon dioxide is then absorbed by plants in the process of **photosynthesis**. As a result, a proportion of the carbon that makes up any plant will be the radioactive form, carbon-14. Included in plant material, the radioactive isotope carbon-14 enters the food chain, which means that animals and humans will also have a proportion of carbon-14 in their bodies. These carbon-14 atoms will decay but, in living plants and animals, they are continuously replaced by new ones.

When an organism (living thing) dies, the replacement process stops. As time passes, the radioactive carbon decays and the proportion of radioactive carbon in the remains of the plant or animal, compared with the stable carbon isotope, decreases.

The half-life for the decay of carbon-14 is approximately 5600 years. This means that every 5600 years the proportion of carbon-14 in dead plant and animal material will halve. The amount of carbon-14 present in a sample of dead plant or animal material is found by measuring the activity of the sample. This is compared with the amount of carbon-14 that would have been present when the sample was part of a living organism. From this, it is possible to estimate when the source of the sample died.

EXAMPLE 1

120 g of living wood has a radioactive activity (corrected for background radiation) of 24 Bq. A 120 g sample of wood from an historical site is found to have an activity of 6 Bq. If the half-life of carbon-14 is 5600 years, estimate the age of the wood from the site.

$$6 \text{ Bq} = 24 \text{ Bq} \times \frac{1}{2} \times \frac{1}{2}$$

Since the activity of the sample has halved twice from that expected in living wood, two half-lives must have passed.

The age of the sample is therefore around 2 × 5600 years = 11 200 years.

There are limitations to the method of radiocarbon dating. It assumes the level of cosmic radiation reaching the Earth is constant, which is not necessarily accurate. Fortunately, the technique has been adjusted to take the variations of cosmic ray activity into account. This is done by testing samples of a known age, like material from the mummies (preserved corpses) of Egyptian Pharaohs and from very ancient living trees.

The radiocarbon method is not used to date samples older than 50 000–60 000 years because, after 10 half-lives, the amount of carbon-14 remaining in samples is too small to measure accurately.

EXTENSION WORK

Radiocarbon dating may be used as a context in questions about the half-life of radioactive isotopes. What follows on dating the age of rocks is included for interest only.

DATING THE AGE OF ROCKS

Inorganic, non-living matter does not absorb carbon-14, so different techniques must be used for finding out the age of rocks and minerals.

When a radioactive substance decays it transforms into a different isotope, sometimes of the same element, sometimes of a different element. The original radioisotope is called the parent nuclide (unstable nucleus) and the product is called the daughter nuclide. Many of the products of decay, the daughter isotopes, are also unstable and these too decay in turn. This means that as the parent isotope decays it breeds a whole family of elements in what is called a decay series. The end of the decay series is a stable isotope – one that does not decay further.

The table shows some radioactive parent isotopes with the stable daughters formed at the end of their particular decay series. The half-life quoted is the time for half the original number of parent nuclei to decay to the stable daughter element.

Radioactive parent isotope	Stable daughter element	Half-life/years
potassium-40	argon-40	1.25 billion
thorium-232	lead-208	14 billion
uranium-235	lead-207	704 billion
uranium-238	lead-206	4.47 billion
carbon-14	nitrogen-14	5568

For rocks containing such radioactive isotopes, the proportion of parent to stable daughter nuclide gives a measure of the age of the rock. Notice that the half-lives of most of the radioactive parent isotopes are extremely long, in some cases greater than the lifetime of the Earth.

The decay series of potassium-40 ends with argon gas. As potassium-40 decays in igneous rock, the argon produced remains trapped in the rock. Igneous rocks are formed when molten rock becomes solid. Igneous rocks are **non-porous**. The proportion of argon to potassium-40 again gives a measure of the age of the rock.

THE DANGERS TO HEALTH OF IONISING RADIATION

Ionising radiation can damage the molecules that make up the cells of living tissue. Cells suffer this kind of damage all the time for many different reasons. Fortunately, cells can repair or replace themselves given time so, usually, no permanent damage results. However, if cells suffer repeated damage because of ionising radiation, the cell may be killed. Alternatively the cell may start to behave in an unexpected way because it has been damaged. We call this effect cell mutation. Some types of cancer happen because damaged cells start to divide uncontrollably and no longer perform their correct function.

Different types of ionising radiation present different risks. Alpha particles have the greatest ionising effect, but they cannot pass through many materials. This means that an alpha source presents little risk, as alpha particles do not penetrate (pass through) the skin. The problem of alpha radiation is much greater if the source of alpha particles is taken into the body. Here the radiation will be very close to many different types of cells and they may be damaged if the exposure is prolonged. Alpha emitters can be breathed in or taken in through eating food. Radon gas is a decay product of radium and is an alpha emitter. It therefore presents a real risk to health. Smokers greatly increase their exposure to this kind of damage as they draw the radiation source right into their lungs (cigarette smoke contains radon).

Beta and gamma radiation do provide a serious health risk when outside the body. Both can penetrate skin and flesh (body) and can cause cell damage by ionisation. Gamma radiation, as we have mentioned earlier, is the most penetrating. The damage caused by gamma rays will depend on how much of their energy is absorbed by ionising atoms along their path. Beta and gamma emitters that are absorbed by the body present less risk than alpha emitters, because of their lower ionising power.

In all cases, the longer the period of exposure to radiation the greater the risk of serious cell damage. Workers in the nuclear industry wear badges to indicate their level of exposure. Some are pieces of photographic film that become increasingly 'foggy' (unclear) as the radiation exposure increases. Another type of badge uses a property called thermoluminescence. Thermoluminescence means that the exposed material will give out light when it is warmed. The radiation releases energy to make heat so the thermoluminescent badges give out more light when exposed to higher levels of radiation. Workers have their badges checked regularly and this gives a measure of their overall exposure to radiation.

SAFE HANDLING OF RADIOACTIVE MATERIALS

Samples of radioactive isotopes used in schools and colleges are very small. This is to limit the risk to users, particularly those who use them regularly – the teachers! Although the risk is small, certain precautions must be followed. The samples are stored in lead-lined containers to block even the most penetrating form of radiation, gamma rays. The containers are clearly labelled as a radiation hazard (danger) and must be stored in a locked metal cabinet. The samples are handled using tongs and are kept as far from the body as possible.

In the nuclear industry and research laboratories, much larger amounts of radioactive material are used. These have to be handled with great care. Very energetic sources will be handled remotely by operators who are protected by lead shields, concrete and thick glass viewing panels. As has been mentioned earlier, neutron radiation is absorbed by lighter elements and waste materials, like spent uranium fuel rods from nuclear reactors, are stored under water until the neutron radiation levels drop to a safe level.



▲ Figure 24.9 Radioactive samples are stored in lead-lined containers and are handled with tongs or protected fingers.



▲ Figure 24.10 Industrial sources of radioactivity must be handed with a lot of care.

The major problem with nuclear materials is long-term storage. Some materials have extremely long half-lives so they remain active for thousands and sometimes tens of thousands of years. Nuclear waste must be stored in sealed containers that must be capable of containing the radioactivity for enormously long periods of time.

CHAPTER QUESTIONS

More questions on applications of radioactivity can be found at the end of Unit 7 on page 255.

SKILLS REASONING

9

1 The most widely used isotope in medicine is technetium-99m. It has a half-life of six hours and decays by the emission of low-energy gamma rays and beta particles.

a Explain why the characteristics of technetium-99m make it suitable for diagnostic use in medicine.

SKILLS CREATIVITY, REASONING

b Technetium-99m can be chemically attached to a wide variety of pharmaceutical products so that it can be targeted at particular tissues or organs. How can its progress through the body be measured and monitored?

SKILLS REASONING

10

2 Technetium-99m is produced from molybdenum-99 in a device called a technetium generator. Which decay process – α, β or γ – could cause molybdenum-99 to decay to technetium-99m? Explain your answer.

HINT

Look at Chapter 22.

SKILLS CREATIVITY, REASONING

8

3 A radioactive isotope of iodine is used in both the diagnosis and treatment of a condition of the thyroid gland. This gland naturally takes up ordinary iodine as part of its function. If a patient has an overactive thyroid it concentrates too much iodine in the gland and this has serious effects on the patient's health.

How might the radioisotope iodine-131 be used to:

a identify an overactive thyroid gland?

b treat the overactive thyroid?

(Iodine-131 has a half-life of eight days and is a high-energy beta emitter.)

SKILLS CRITICAL THINKING

4 a Explain the difference between radioactive contamination and irradiation.

b Give an example of a use of irradiation.

SKILLS CREATIVITY, REASONING

5 Paper is made in a variety of different 'weights', with different thicknesses. How could ionising radiation be used to check the thickness of paper during production? You should consider the following:

- the type of radiation to be used
- how it will be used to measure the paper thickness
- what checks should be made to ensure that the measurements are accurate
- safety procedures.

SKILLS CRITICAL THINKING

6 Radiocarbon dating is used to estimate the age of organic (once-living) materials. It uses a radioisotope of carbon, carbon-14 (C-14).

a How is carbon-14 formed?

b Why does all living matter contain a proportion of C-14?

c What happens to the proportion of C-14 in an organism once it has died?

d What assumptions are made in the process of radiocarbon dating?

e Why is this method unsuitable for accurately dating material that is more than 50 000 years old?

SKILLS PROBLEM SOLVING

7 Most radioactive isotopes of elements have half-lives that are extremely short compared to the age of the Earth. The Earth is about 4.5 billion years old. Radium has a half-life of only about 1600 years.

a How many half-lives of radium have there been since the Earth formed?

SKILLS ANALYSIS

Student A says all the radium formed when the Earth condensed out of the Sun's atmosphere, should have decayed away to an unmeasurably small amount by now.

Student B says that it depends on how much there was to start with.

Student C says that there is still a significant amount of radium on the Earth.

b Which student or students are correct? Give reasons for your answer.

SKILLS REASONING

8 An isotope that decays by alpha emission is relatively safe when outside the body but very dangerous if absorbed by the body, either through breathing or eating.

a Explain why this is so.

b Why is radon-220 a particularly dangerous isotope?